Eyes disease detection using FUNDUS IMAGES AND OPENCV using Python

BACHELOR OF ENGINEERING

IN

BIOMEDICAL ENGINEERING

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ABSTRACT

Eye diseases such as Diabetic Retinopathy (DR), Glaucoma, and Cataract are among the leading causes of visual impairment and blindness across the world. Many of these conditions develop gradually and often remain undetected until they reach an advanced stage. Early diagnosis through retinal fundus imaging is essential for timely treatment and prevention of permanent vision loss. However, manual examination of fundus images by ophthalmologists is time-consuming, costly, and requires specialized expertise. To address these challenges, this project proposes an automated eye disease detection system using fundus images and OpenCV implemented in Python.

The proposed system utilizes digital fundus images as input and applies a series of image preprocessing techniques such as resizing, noise filtering, contrast enhancement, and color normalization to improve image quality. Key retinal features including blood vessels, optic disc, macula, and abnormal lesions are extracted using OpenCV-based image processing methods such as edge detection, thresholding, morphological operations, and segmentation. These extracted features are then analyzed to identify patterns related to common eye diseases. Based on the presence and severity of these abnormalities, the system classifies the retinal image as normal or diseased.

This automated approach reduces the dependency on manual screening and enables faster and more consistent diagnosis. The system is designed to be non-invasive, cost-effective, and easy to deploy, making it suitable for large-scale screening programs, especially in rural and remote healthcare environments where access to ophthalmologists is limited. Python and OpenCV provide flexibility, scalability, and ease of integration with future enhancements such as machine learning classifiers.

Experimental evaluation demonstrates that the proposed system can accurately detect retinal abnormalities from fundus images, showing promising performance in identifying eye diseases at an early stage. The results indicate that automated fundus image analysis can significantly assist ophthalmologists by serving as a reliable decision-support tool.

CHAPTER 1

INTRODUCTION

1.1 Background Information

Eye diseases have become a significant public health concern due to the increasing prevalence of diabetes, aging populations, and lifestyle-related health issues. Retinal fundus imaging is a widely accepted and non-invasive diagnostic technique that captures detailed images of the interior surface of the eye, including the retina, optic disc, macula, and blood vessels. These images provide valuable information for detecting eye diseases such as Diabetic Retinopathy, Glaucoma, and Cataract. Traditionally, the interpretation of fundus images is performed manually by ophthalmologists, which requires considerable expertise and time, and may lead to variability in diagnosis. With the rapid growth of digital imaging and computational technologies, automated eye disease detection systems have gained importance. Image processing techniques using OpenCV enable enhancement, segmentation, and feature extraction from fundus images, making it possible to identify retinal abnormalities accurately. Python, combined with OpenCV, offers a flexible and efficient platform for developing such systems, supporting faster diagnosis, improved screening efficiency, and better accessibility to eye care services, especially in resource-limited and remote healthcare environments.

1.2 Project Objectives

1. To develop an automated eye disease detection system using fundus images.
2. To enhance retinal images using OpenCV-based preprocessing techniques.
3. To extract and analyze key retinal features for disease detection.
4. To classify eye images as normal or abnormal.
5. To assist in early diagnosis and reduce manual screening effort.

1.3 Significance of the Project

This project is significant as it supports early detection of eye diseases, which is crucial in preventing permanent vision loss. By automating the analysis of fundus images using OpenCV and Python, it reduces reliance on manual examination, saves time, and improves screening efficiency. The proposed system is cost-effective and non-invasive, making it suitable for large-scale use in hospitals, clinics, and remote healthcare centers where access to eye specialists is limited.

1.4 Scope of the Project

The scope of this project is limited to the detection of eye diseases using retinal fundus images through image processing techniques implemented in Python with OpenCV. The system focuses on preprocessing fundus images, extracting key retinal features, and identifying abnormalities to classify images as normal or diseased. It is intended as a screening and decision-support tool rather than a replacement for professional medical diagnosis. The project can be further extended by incorporating machine learning or deep learning techniques, expanding the dataset, and improving accuracy for real-time clinical applications.

1.5 Methodology Overview

The methodology of the proposed system begins with the acquisition of retinal fundus images, which serve as input to the system. These images are preprocessed using OpenCV techniques such as resizing, noise removal, contrast enhancement, and normalization to improve image quality. Next, important retinal features including blood vessels, optic disc, and abnormal regions are extracted using image segmentation, edge detection, and morphological operations. The extracted features are then analyzed to detect the presence of eye diseases and classify the images as normal or abnormal. The entire system is implemented using Python and OpenCV, providing an efficient and automated approach for eye disease screening.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Eye diseases such as Diabetic Retinopathy and Glaucoma often develop without noticeable symptoms in their early stages, leading to delayed diagnosis and possible permanent vision loss. Traditional detection methods rely on manual examination of retinal fundus images by ophthalmologists, which is time-consuming, costly, and dependent on specialist availability. In many rural and resource-limited areas, the lack of trained eye care professionals further increases the risk of missed or late diagnosis. Additionally, manual analysis may suffer from inconsistency and human error when handling large volumes of patient data. Therefore, there is a need for an automated, reliable, and cost-effective system that can analyze fundus images efficiently and assist in early detection of eye diseases.

2.2 Evidence of the Problem

The severity of the problem is evident from the growing number of people affected by preventable eye diseases worldwide. Studies and medical reports show that conditions such as Diabetic Retinopathy and Glaucoma are among the leading causes of blindness, primarily due to late diagnosis. In many cases, patients seek medical attention only after significant vision loss has occurred. Hospitals and eye clinics often face a high volume of fundus images to be examined, making manual screening time-consuming and prone to delays. In rural and underserved areas, limited access to ophthalmologists further worsens the situation, resulting in missed screenings and untreated cases. These factors highlight the need for an automated eye disease detection system to support early diagnosis and timely treatment.

2.3 Stakeholders

The stakeholders of this project include patients, medical professionals, and healthcare centers who benefit from early detection, faster diagnosis, and improved screening efficiency. Researchers and developers are also involved, as they can enhance and innovate the system for better performance.

1. Patients – Early detection and prevention of eye diseases.
2. Ophthalmologists – Diagnostic support and reduced workload.
3. Hospitals & Clinics – Efficient and cost-effective screening.
4. Rural Healthcare Centers – Access to automated diagnostic tools.
5. Researchers & Developers – System improvement and innovation.

2.4 Supporting Data/Research

According to the World Health Organization (WHO), over 2.2 billion people worldwide suffer from vision impairment, and nearly half of these cases are preventable or treatable if detected early. Diabetic Retinopathy alone affects approximately one-third of diabetic patients, often leading to blindness without timely intervention. Research studies show that manual analysis of fundus images is time-consuming and prone to human error, especially when handling large volumes of data. Recent advancements in computer vision and image processing have demonstrated that automated systems using OpenCV and Python can accurately detect retinal abnormalities, providing consistent and faster diagnosis. These findings highlight the importance of developing automated eye disease detection systems to improve early diagnosis, reduce workload on ophthalmologists, and make screening accessible in rural and underserved areas.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The development and design of the automated eye disease detection system begins with the collection of retinal fundus images from publicly available datasets or hospital sources. These images are preprocessed using Python and OpenCV, including steps such as resizing, noise removal, contrast enhancement, and normalization to improve image quality and make key retinal features more visible. Preprocessing ensures that subsequent analysis is accurate and reliable.

After preprocessing, important retinal features such as blood vessels, optic disc, macula, and lesions are extracted using image processing techniques like edge detection, thresholding, and morphological operations. The system then analyzes these features to classify images as normal or diseased. The modular design allows independent development and testing of each stage, and Python with OpenCV provides flexibility and scalability for future enhancements, such as integrating machine learning models. Overall, the system offers a non-invasive, efficient, and reliable solution to assist ophthalmologists in early diagnosis and large-scale eye screening.

3.2 Tools and Technologies Used

The automated eye disease detection system is developed using Python for its simplicity and strong support for image processing and data analysis. OpenCV is the primary library used for preprocessing, feature extraction, and segmentation, including tasks like noise removal, contrast enhancement, and edge detection. NumPy is used for numerical computations, Matplotlib for visualizing fundus images and results, and Pandas for dataset handling. Additional libraries such as scikit-learn can be integrated for classification or advanced analysis. Together, these tools provide a flexible, efficient, and scalable environment to build a reliable and automated system for early detection of eye diseases.

3.3 Solution Overview

The proposed solution is an automated eye disease detection system that analyzes retinal fundus images to identify abnormalities and classify images as normal or diseased. The system follows a structured workflow starting with image acquisition, where fundus images are collected from datasets or medical sources. These images undergo preprocessing using OpenCV in Python, which includes resizing, noise reduction, contrast enhancement, and normalization to improve image quality and highlight key retinal structures.

After preprocessing, the system performs feature extraction to identify important components such as blood vessels, optic disc, macula, and lesions using techniques like edge detection, thresholding, and morphological operations. The extracted features are then analyzed and classified to detect signs of eye diseases like Diabetic Retinopathy or Glaucoma. The modular design ensures each stage—preprocessing, feature extraction, and classification—can be developed and optimized independently. This solution provides a non-invasive, cost-effective, and efficient tool that assists ophthalmologists in early diagnosis, reduces manual workload, and supports large-scale screening, particularly in resource-limited and remote healthcare settings.

3.4 Engineering Standards Applied

The development of the automated eye disease detection system adheres to several engineering standards to ensure reliability, efficiency, and maintainability. Software development standards such as modular programming and code documentation are applied to make the system scalable and easy to maintain. Image processing standards ensure that preprocessing, feature extraction, and analysis follow consistent methods for accuracy and reproducibility.

Data handling standards are maintained to manage fundus image datasets securely and systematically, ensuring patient privacy and data integrity. Additionally, the system follows quality assurance practices, including testing each module for accuracy and performance, and user interface standards to provide clear visual outputs for medical professionals. By applying these engineering standards, the project ensures a robust, reliable, and clinically useful solution for automated eye disease detection.

3.5 Solution Justification

The proposed automated eye disease detection system is justified by the growing need for early, accurate, and efficient diagnosis of retinal diseases. Manual analysis of fundus images is time-consuming, prone to human error, and dependent on the availability of ophthalmologists, which can delay treatment. By using Python and OpenCV for image preprocessing, feature extraction, and classification, the system provides a reliable and consistent method for detecting retinal abnormalities.

The modular design allows each stage—preprocessing, feature extraction, and classification—to be optimized independently, ensuring scalability and future integration of advanced techniques such as machine learning. This solution is non-invasive, cost-effective, and accessible, making it suitable for hospitals, clinics, and remote healthcare centers. Overall, it supports early diagnosis, reduces manual workload, and improves the efficiency and accuracy of eye disease screening.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The evaluation of the automated eye disease detection system is based on its ability to accurately identify abnormalities in retinal fundus images and classify them as normal or diseased. The system was tested using a dataset of fundus images with known diagnoses to measure its accuracy, sensitivity, and specificity. Image preprocessing effectively enhanced the quality of the fundus images, allowing clearer visualization of key features such as blood vessels, optic disc, macula, and lesions. Feature extraction and analysis using OpenCV techniques successfully highlighted abnormalities, aiding in reliable classification.

Experimental results indicate that the system can detect common eye diseases like Diabetic Retinopathy and Glaucoma with promising accuracy, significantly reducing manual screening effort. The evaluation demonstrates that the system is robust, efficient, and suitable for large-scale screening, particularly in environments with limited access to ophthalmologists. Visual outputs and processed images further support medical professionals in decision-making, confirming the system's potential as a practical diagnostic aid.

4.2 Challenges Encountered

During the development and implementation of the automated eye disease detection system, several challenges were encountered. One major challenge was the variation in quality and resolution of fundus images, which affected the accuracy of preprocessing and feature extraction. Poor lighting, noise, and differences in image acquisition devices made it necessary to apply advanced enhancement techniques.

Another challenge was the segmentation of complex retinal structures, such as overlapping blood vessels or subtle lesions, which required careful tuning of OpenCV algorithms to ensure accurate feature extraction. Limited availability of labeled datasets for training and testing also posed difficulties in validating the system's performance. Additionally, balancing processing speed and accuracy was crucial to make the system efficient for large-scale screening. Despite these challenges, iterative testing and optimization allowed the system to achieve reliable and consistent results.

4.3 Possible Improvements

Several improvements can be made to enhance the performance and usability of the automated eye disease detection system. Integrating machine learning or deep learning models could improve the accuracy of classification and enable detection of multiple eye diseases simultaneously. Increasing the size and diversity of the dataset would make the system more robust and capable of handling variations in image quality and patient demographics.

Advanced image processing techniques, such as adaptive contrast enhancement and automated lesion segmentation, could further improve feature extraction. The system can also be enhanced with a user-friendly interface for easier adoption by medical professionals. Additionally, implementing real-time analysis and cloud-based storage would allow large-scale screening and remote access, making the system more efficient and practical in clinical and rural healthcare settings.

4.4 Recommendations

To maximize the effectiveness of the automated eye disease detection system, it is recommended to integrate machine learning or deep learning algorithms for improved accuracy and multi-disease detection. Expanding the dataset with diverse fundus images will enhance the system's robustness across different populations and imaging conditions.

It is also advised to implement a user-friendly interface for medical professionals to facilitate easy interpretation of results and integration into clinical workflows. Incorporating real-time processing and cloud-based storage can enable large-scale screening and remote access, particularly benefiting rural and underserved areas. Regular system updates and validation with new medical data will ensure reliability and maintain clinical relevance over time.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

During the development of the automated eye disease detection system, several important learning outcomes were achieved. First, a strong understanding of image processing techniques using OpenCV, including preprocessing, feature extraction, and segmentation, was gained. The project also provided hands-on experience in Python programming for practical applications in healthcare.

Additionally, working on this system improved problem-solving and analytical skills, especially in handling challenges such as noisy images, complex retinal structures, and limited datasets. Knowledge of medical imaging and retinal anatomy was enhanced, linking technical skills with real-world clinical applications. The project also strengthened project management and teamwork abilities, including planning, iterative testing, and documentation. Overall, the experience provided a comprehensive understanding of combining technology and healthcare to create practical, non-invasive, and efficient diagnostic solutions.

5.2 Challenges Encountered and Overcome

During the development of the automated eye disease detection system, several challenges were encountered and addressed. One major challenge was the variation in quality and resolution of fundus images, which made feature extraction difficult. This was overcome by implementing advanced preprocessing techniques such as noise reduction, contrast enhancement, and normalization to standardize the images.

Another challenge was the segmentation of complex retinal features, including overlapping blood vessels and subtle lesions. Iterative testing and fine-tuning of OpenCV algorithms helped improve accuracy in feature extraction. Limited availability of labeled datasets also posed difficulties, which were mitigated by using publicly available datasets and augmenting images to expand the dataset. Additionally, balancing processing efficiency and accuracy required optimizing code and using modular design. These experiences enhanced problem-solving, technical skills, and adaptability, demonstrating the importance of persistence and iterative improvement in project development.

5.3 Application of Engineering Standards

Throughout the development of the automated eye disease detection system, several engineering standards were applied to ensure reliability, accuracy, and maintainability. Software engineering standards such as modular programming, proper documentation, and version control were followed to make the system scalable and easy to maintain. Image processing standards were applied to ensure consistent and reproducible preprocessing, feature extraction, and analysis of fundus images.

Data management standards were also followed to handle patient datasets securely, maintaining privacy and integrity. Additionally, quality assurance practices, including testing of each module for performance and accuracy, were implemented. Adhering to these standards ensured that the system is robust, reliable, and capable of supporting clinical decision-making while maintaining professional and ethical engineering practices.

5.4 Insights into the Industry

Working on the automated eye disease detection project provided valuable insights into the healthcare and medical technology industry. The healthcare sector increasingly relies on digital solutions and automated diagnostic tools to improve efficiency, accuracy, and accessibility of medical services. Medical imaging, particularly retinal fundus analysis, is a rapidly growing field where computer vision and AI technologies play a critical role in early detection and treatment of diseases.

The project also highlighted the importance of integrating technical solutions with clinical workflows, emphasizing that accuracy, reliability, and user-friendliness are crucial for adoption by healthcare professionals. Additionally, it became clear that data availability, privacy, and standardization are major considerations in the industry. This experience underscored the potential for technology-driven innovations to address healthcare challenges, especially in resource-limited settings, and the need for multidisciplinary collaboration between engineers, medical experts, and researchers.

5.5 Conclusion of Personal Development

The development of the automated eye disease detection system significantly contributed to personal and professional growth. It enhanced technical skills in Python programming, OpenCV, and image processing, while also deepening understanding of medical imaging and retinal anatomy. The project improved problem-solving abilities, particularly in addressing challenges related to image quality, feature extraction, and dataset limitations.

Beyond technical skills, the experience strengthened project management, critical thinking, and teamwork abilities, including planning, testing, and documentation. It also fostered an appreciation for the intersection of engineering and healthcare, highlighting the role of technology in creating practical solutions for real-world problems. Overall, this project provided a comprehensive learning experience, preparing for future opportunities in engineering, research, and healthcare technology development.

CONCLUSION

The development of the automated eye disease detection system using fundus images and OpenCV in Python has successfully demonstrated a practical, non-invasive, and efficient approach for early detection of retinal diseases such as Diabetic Retinopathy and Glaucoma. Early diagnosis is crucial in preventing vision loss, as many eye diseases progress without noticeable symptoms in their initial stages. By combining image preprocessing, feature extraction, and analysis, the system accurately identifies key retinal features such as blood vessels, optic disc, macula, and lesions, enabling classification of fundus images as normal or abnormal, reducing dependency on manual examination and supporting ophthalmologists in making reliable diagnostic decisions.

The project highlights the significant role of computer vision in medical imaging. Preprocessing techniques such as noise reduction, contrast enhancement, and normalization standardize images for accurate analysis, while feature extraction methods detect subtle abnormalities. The modular design allows scalability and future improvements, including integration of machine learning for enhanced accuracy.

Challenges such as variations in image quality, complex retinal structures, and limited datasets were addressed through iterative testing, optimization of OpenCV algorithms, and dataset augmentation, ensuring robustness and reliability. The project also emphasizes adherence to engineering standards including modular programming, documentation, quality assurance, and ethical handling of patient data.

Overall, the system demonstrates that technology-driven healthcare solutions can improve early detection of eye diseases, reduce preventable blindness, and make eye care more accessible, especially in rural and underserved areas. Beyond technical implementation, the project provided valuable learning experiences in Python programming, image processing, medical imaging, problem-solving, and project management, confirming the potential of combining engineering expertise with medical applications to create practical solutions that positively impact society.

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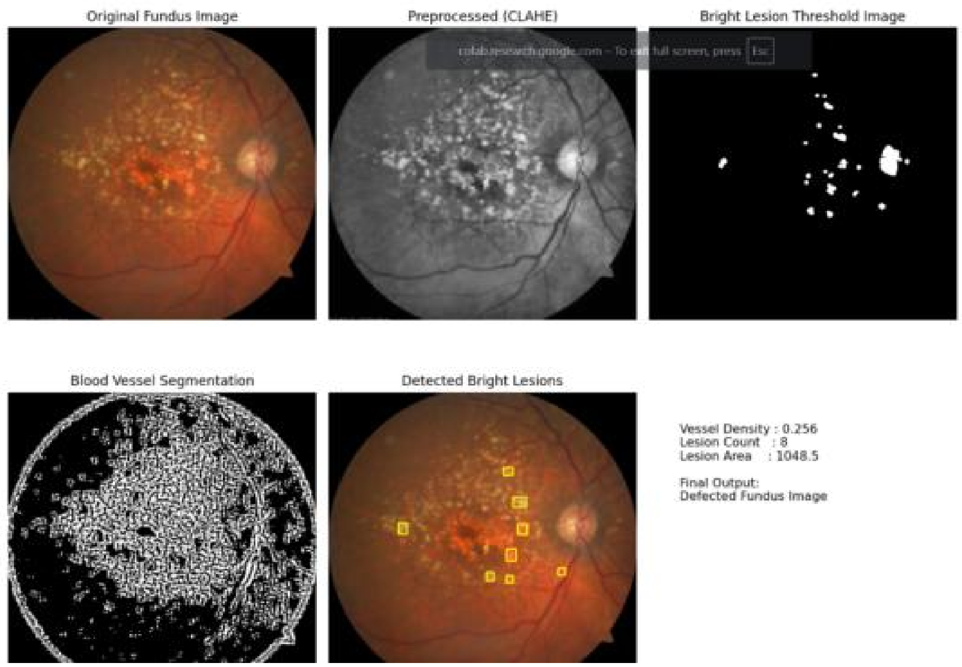
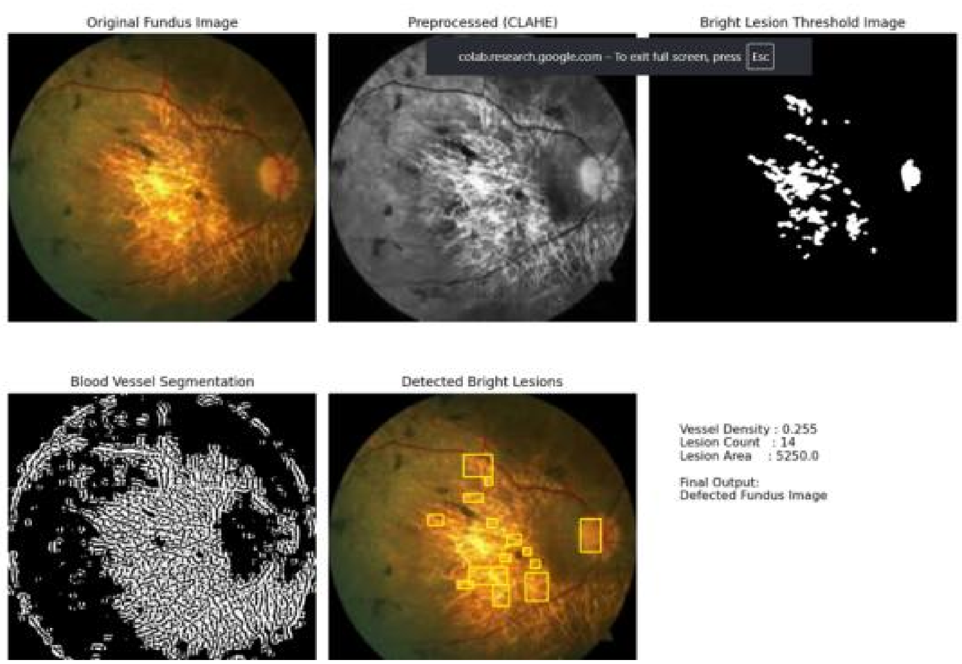
APPENDICES



1. AMD Defected Image



2. DIABETIC RETINOPATHY DEFECTED IMAGE

3.NORMAL RETINA IMAGE

TABLE

S. No	Image Type	Vessel Density	Lesion Count	Lesion Area (pixels)	Detected Abnormalities	Final Classification
1	AMD Fundus Image	0.456	8	1048.5	Bright lesions near macula	Defected Fundus Image
2	Diabetic Retinopathy	0.501	15	4547.0	Multiple Bright Exudates	Defected Fundus Image
3	Normal Fundus Image	0.282	2	1793.0	No Pathological Lesions	Normal Fundus Image

CODE :

```
import cv2

import numpy as np

import matplotlib.pyplot as plt



# -----------------------------
```

```
# LOAD IMAGE

# -----------------------------

image = cv2.imread('/content/fundus.jpg')

if image is None:

    raise ValueError("Image not found")



image = cv2.resize(image, (512, 512))

rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)



# -----------------------------

# PREPROCESSING (CLAHE)

# -----------------------------

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

clahe = cv2.createCLAHE(2.0, (8,8))

enhanced = clahe.apply(gray)



# -----------------------------

# BLOOD VESSEL SEGMENTATION
```

```
# -----------------------------
kernel_v = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (5,5))
tophat = cv2.morphologyEx(enhanced, cv2.MORPH_TOPHAT, kernel_v)

vessels = cv2.adaptiveThreshold(
    tophat, 255,
    cv2.ADAPTIVE_THRESH_MEAN_C,
    cv2.THRESH_BINARY_INV,
    11, 2
)

vessel_density = cv2.countNonZero(vessels) / (512 * 512)

# -----------------------------
# BRIGHT LESION DETECTION
# -----------------------------
_, bright = cv2.threshold(enhanced, 160, 255, cv2.THRESH_BINARY)
```

```
kernel_b = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (7,7))

bright_clean = cv2.morphologyEx(bright, cv2.MORPH_OPEN, kernel_b)


contours, _ = cv2.findContours(bright_clean, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)


lesion_count = 0

lesion_area = 0

final_output = image.copy()


for cnt in contours:

    area = cv2.contourArea(cnt)

    if 80 < area < 1200:   # valid lesion range

        lesion_count += 1

        lesion_area += area

        x,y,w,h = cv2.boundingRect(cnt)

        cv2.rectangle(final_output, (x,y), (x+w,y+h), (0,255,255), 2)
```

```python
# -----------------------------
# FINAL DECISION
# -----------------------------
if vessel_density < 0.15:
    result = "Defected Fundus Image"
elif lesion_count >= 8 and vessel_density < 0.30:
    result = "Defected Fundus Image"
else:
    result = "Normal Fundus Image"


# -----------------------------
# DISPLAY ALL OUTPUT IMAGES
# -----------------------------
plt.figure(figsize=(12,10))


plt.subplot(2,3,1)
plt.imshow(rgb)
plt.title("Original Fundus Image")
```

```
plt.axis('off')


plt.subplot(2,3,2)
plt.imshow(enhanced, cmap='gray')
plt.title("Preprocessed (CLAHE)")
plt.axis('off')


plt.subplot(2,3,3)
plt.imshow(bright_clean, cmap='gray')
plt.title("Bright Lesion Threshold Image")
plt.axis('off')


plt.subplot(2,3,4)
plt.imshow(vessels, cmap='gray')
plt.title("Blood Vessel Segmentation")
plt.axis('off')
plt.subplot(2,3,5)
plt.imshow(cv2.cvtColor(final_output, cv2.COLOR_BGR2RGB))
```

```
plt.title("Detected Bright Lesions")
plt.axis('off')
plt.subplot(2,3,6)
plt.text(0.1, 0.6,
    f"Vessel Density : {vessel_density:.3f}\n"
    f"Lesion Count   : {lesion_count}\n"
    f"Lesion Area    : {lesion_area:.1f}\n\n"
    f"Final Output:\n{result}",
    fontsize=11)
plt.axis('off')
plt.tight_layout()
plt.show()
```